Lab 14

Preparing the Mini-Project Demonstration

Communication Technologies for the Internet of Things | Lecture 14

Aim and context

The mini-project demonstration is oral and takes place in class. There is no report: the working system is the document. This guide helps each group prepare the demonstration, with a checklist based on the assignment, a rehearsal with induced failures and a traffic capture that supports a claim.

The guide is not an assignment to hand in. It is a working outline for the group, to be used before class.

Learning outcomes

- Describe the system architecture in two minutes, with a diagram.
- Verify each aspect requested in the assignment, with evidence.
- Induce failures and show the system's reaction.
- Capture traffic and use it to support a claim.
- Declare and explain the use of AI.

How to use this guide

Each step presents, in this order, the aim, the context, the theoretical background, the commands to run and the expected result. Read the whole step before running the first command. The boxes always have the same meaning:

Command to run

```
Command to run in the terminal.
```

Expected result (values may vary)

```
Result that should be displayed.
```

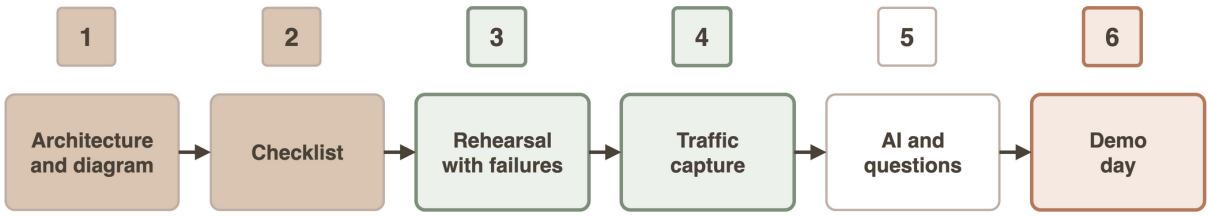
Analyse

A moment to pause and interpret the result, on your own or in discussion with colleagues.

Possible error and fix

An error that may occur and how to fix it.

Overview



Preparing the demonstration means rehearsing it against the assignment's checklist

How to read the figure

The six steps read from left to right. The first two, in beige, fix what is going to be shown. Steps 3 and 4, in green, are the rehearsal with failures and the traffic capture. Step 5 deals with AI and questions, and step 6 with demo day.

Step 1 — Architecture and diagram

Aim. Have a diagram and a two-minute explanation of the architecture.

Theoretical background

The mini-project's reference architecture has four parts: acquisition node, gateway, middleware and consumers. Each theme adds details, but the explanation always starts with the path of the data, from sensor to consumer. The presenter should name, for each arrow, the protocol and the message format.

Analyse

Can you say, without looking, which protocol and which format each link of your system uses? What is the topic, or the resource, of each message?

Step 2 — Checklist

Aim. Verify each aspect of the assignment and identify the evidence that demonstrates it.

Theoretical background

An aspect only counts if it can be shown. The following list reproduces the aspects of the assignment, and for each there should be a command, a screen or a capture that evidences it.

2.1 Wireless communication

- [ ] Mechanism chosen and justification (range, data rate, consumption).
- [ ] Reconnection when the device fails.
- [ ] Comparison with at least one alternative.

2.2 MQTT broker

- [ ] Coherent and documented topic hierarchy.
- [ ] Subscription with wildcards.
- [ ] QoS justified by message type.
- [ ] Retained messages.
- [ ] Last Will and Testament.
- [ ] Keep-alive and behaviour on connection loss.
- [ ] Authentication, with no secrets in the code.
- [ ] At least two independent subscribers on the same data.

2.3 CoAP

- [ ] Resources defined and reachable by GET.
- [ ] Confirmable and non-confirmable messages.

- [ ] Resource observation.
- [ ] Comparison with MQTT and HTTP (signalling, size, interaction model).

2.4 Consumers and operation

- [ ] Analytics: aggregation over the day with visualisation.
- [ ] Alarms: threshold rule and notification.
- [ ] Serialisation: data saved to a file and read back.
- [ ] Energy or bandwidth saving.
- [ ] Command channel from the broker to the actuator.
- [ ] Behaviour under failures (device, gateway, broker).
- [ ] Traffic capture that supports a claim.

Step 3 — Rehearsal with induced failures

Aim. Induce failures and confirm that the system reacts as claimed.

Theoretical background

A demonstration that only works when everything goes well does not show robustness. Inducing the failure of a node, the gateway or the broker, and showing the reaction, is the most direct way to evidence the last will, keep-alive, reconnection and message retention.

In a terminal, watch everything that passes through the broker during the rehearsal:

Command to run

```
mosquitto_sub -h localhost -t '#' -v
```

In another terminal, stop and restart the component to be tested, for example the broker:

Command to run

```
sudo systemctl stop mosquitto
sleep 10
sudo systemctl start mosquitto
```

Analyse

What happened to the gateway and the consumers when the broker stopped? Did they reconnect on their own? What was lost and what was kept? How do you justify it with the QoS and keep-alive chosen?

Step 4 — Traffic capture

Aim. Obtain a capture that supports a claim made in the demonstration.

Theoretical background

A claim such as “MQTT uses fewer octets than HTTP” or “this message is sent with QoS 1” is only solid with evidence. `tcpdump` captures packets on the command line, and Wireshark presents them decoded by protocol. The `mqtt` and `coap` filters show only the relevant traffic.

Command to run

```
sudo tcpdump -i any -w demo.pcap 'port 1883 or udp port 5683'
```

Open the `demo.pcap` file in Wireshark and use the filter `mqtt.msgtype == 3` to see PUBLISH messages, and the `mqtt.qos` field to confirm the QoS level. For CoAP, use the `coap` filter.

Analyse

Which claim in your demonstration is supported by this capture? What is the size of a PUBLISH message with your payload, and what fraction is header?

Step 5 — AI declaration and questions

Aim. Prepare the AI declaration and anticipate questions.

Theoretical background

The use of AI is permitted, must be declared and is critically assessed. A good declaration says which tool was used, for what, what was accepted, what was corrected and how it was verified. All group members must be able to explain any part of what is presented.

- ☐ Each group member can explain the code they present.
- ☐ There is a list of the points where AI contributed and what was corrected.
- ☐ The group anticipated questions about the choices of QoS, topics and wireless technology.

Step 6 — Demo day

Aim. Arrive with everything ready and with a plan B.

- [ ] Hardware connected and tested before class, with copies of configuration files without secrets.
- [ ] Plan B for network and hardware failure, for example a recording of the system working.
- [ ] Rehearsed sequence: architecture, system working, induced failure, traffic capture, AI declaration.
- [ ] Timed, with margin for questions.

Summary

- Preparing is rehearsing against the **checklist**, with evidence for each aspect.
- An **induced failure** shows robustness better than a perfect demonstration.
- A **traffic capture** turns a claim into evidence.
- **AI** is declared and explained, and every member can answer for everything.